GATE CSE: 1. Discrete Maths - 2. Set Theory and Algebra

1. **[GATE 1995 — 1 Mark]**

   Let R be a symmetric and transitive relation on a set A. Then

   A. R is reflexive and hence an equivalence relation

   B. R is reflexive and hence a partial order

   C. R is reflexive and hence not an equivalence relation

   D. None of the above

2. **[GATE 1995 — 1 Mark]**

The number of elements in the power set P(S) of the set $S = \{(\phi), 1, (2, 3)\}$ is:

A. 2

B. 4

C. 8

D. None of these

3. **[GATE 1995 — 2 Mark]**

Let A be the set of all nonsingular matrices over real numbers and let $*$ be the matrix multiplication operator. Then

A. A is closed under $*$ but $\langle A, *\rangle$ is not a semigroup.

B. $\langle A, *\rangle$ is a semigroup but not a monoid.

C. $\langle A, *\rangle$ is a monoid but not a group.

D. $\langle A, *\rangle$ is a group but not an abelian group

4. **[GATE 1996 — 1 Mark]**

Let A and B be sets and let A^c and B^c denote the complements of the sets A and B. The set $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to

A. $A \cup B$

B. $A^c \cup B^c$

C. $A \cap B$

D. $A^c \cap B^c$

5. **[GATE 1996 — 1 Mark]**

Let $X = \{2, 3, 6, 12, 24\}$. Let $\leq$ be the partial order defined by $x \leq y$ if x divides y. The number of edges in the Hasse diagram of $(X, \leq)$ is

A. 3

B. 4

C. 9

D. None of these

6. **[GATE 1996 — 1 Mark]**

Suppose X and Y are sets and $|X|$ and $|Y|$ are their respective cardinalities. It is given that there are exactly 97 functions from X to Y. From this one can conclude that

A. $|X| = 1, |Y| = 97$

B. $|X| = 97, |Y| = 1$

C. $|X| = 97, |Y| = 97$

D. None of the above

7. **[GATE 1996 — 1 Mark]**

Which of the following statements is false?

A. The set of rational numbers is an abelian group under addition

B. The set of integers in an abelian group under addition

C. The set of rational numbers form an abelian group under multiplication

D. The set of real numbers excluding zero in an abelian group under multiplication

8. **[GATE 1996 — 2 Mark]**

Let R denote the set of real numbers. Let $f : R \times R \to R \times R$ be a bijective function defined by $f(x, y) = (x + y,\ x - y)$. The inverse function of f is given by

A. $f^{-1}(x, y) = \left(\frac{1}{x+y}, \frac{1}{x-y}\right)$

B. $f^{-1}(x, y) = (x - y,\ x + y)$

C. $f^{-1}(x, y) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$

D. $f^{-1}(x, y) = (2(x - y),\ 2(x + y))$

9. **[GATE 1996 — 2 Mark]**

Let R be a non-empty relation on a collection of sets defined by $A\,R\,B$ if and only if $A \cap B = \phi$. Then, (pick the true statement)

A. R is reflexive and transitive

B. R is symmetric and not transitive

C. R is an equivalence relation

D. R is not reflexive and not symmetric

10. **[GATE 1996 — 2 Mark]**

Which one of the following is false?

A. The set of all bijective functions on a finite set forms a group under function composition.

B. The set $\{1, 2, ..., p-1\}$ forms a group under multiplication mod p where p is a prime number.

C. The set of all strings over a finite alphabet Σ forms a group under concatenation.

D. A subset $s \neq \phi$ of G is a subgroup of the group $\langle G, * \rangle$ if and only if for any pair of elements $a, b \in s$, $a * b^{-1} \in s$.

11. **[GATE 1997 — 1 Mark]**

The number of equivalence relations on the set $\{1, 2, 3, 4\}$ is

A. 15

B. 16

C. 24

D. 4

12. **[GATE 1998 — 1 Mark]**

Suppose A is a finite set with n elements. The number of elements in the Largest equivalence relation of A is

A. n

B. n^2

C. 1

D. $n+1$

13. **[GATE 1998 — 1 Mark]**

Let R_1 and R_2 be two equivalence relations on a set. Consider the following assertions: (i) $R_1 \cup R_2$ is an equivalence relation (ii) $R_1 \cap R_2$ is an equivalence relation Which of the following is correct?

A. both assertions are true

B. assertion (i) is true but assertion (ii) is not true

C. assertion (ii) is true but assertion (i) is not true

D. neither (i) nor (ii) is true

14. **[GATE 1998 — 1 Mark]**

The number of functions from an m element set to an n element set is

A. $m + n$

B. m^n

C. n^m

D. $m * n$

15. **[GATE 1998 — 2 Mark]**

The binary relation $R = (1,1),(2,1),(2,2),(2,3),(2,4),(3,1),(3,2),(3,3),(3,4)$ on the set $A = 1,2,3,4$ is

A. Reflexive, symmetric and transitive

B. Neither reflexive, nor irreflexive but transitive

C. Irreflexive, symmetric and transitive

D. Irreflexive and antisymmetric

16. **[GATE 1998 — 2 Mark]**

Suppose $A = a, b, c, d$ and Π_1 is the following partition of A

$$\Pi_1 = a, b, c, d$$

(a) List the ordered pairs of the equivalence relations induced by Π_1 (b) Draw the graph of the above equivalence relation

Answer: ______________

17. **[GATE 1998 — 2 Mark]**

Let $(A, *)$ be a semigroup. Furthermore, for every a and b in A, if $a \neq b$, then $a * b \neq b * a$. (a) Show that for every a in A

$$a * a = a$$

(b) Show that for every a, b in A

$$a * b * a = a$$

(c) Show that for every a, b, c in A

$$a * b * c = a * c$$

Answer: _______________

18. **[GATE 1999 — 1 Mark]**

The number of binary relations on a set with n elements is

A. n^2

B. $2n$

C. 2^{n^2}

D. None of these

19. **[GATE 1999 — 2 Mark]**

Mr. X claims the following: If a relation R is both symmetric and transitive, then R is reflexive. For this, Mr. X offers the following proof. "From xRy, using symmetry we get yRx. Now because R is transitive, xRy and yRx together imply xRx. Therefore, R is reflexive." Briefly point out the flaw in Mr. X's proof. (b) Give an example of a relation R which is symmetric and transitive but not reflexive.

Answer: ______________

20. **[GATE 2000 — 2 Mark]**

A relation R is defined on the set of integers as xRy iff $(x + y)$ is even. Which of the following statements is true?

A. R is not an equivalence relation

B. R is an equivalence relation having 1 equivalence class

C. R is an equivalence relation having 2 equivalence classes

D. R is an equivalence relation having 3 equivalence classes

21. **[GATE 2000 — 2 Mark]**

Let $P(S)$ denote the power set of a set S. Which of the following is always true?

A. $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\phi\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

22. **[GATE 2001 — 1 Mark]**

Consider the following relations: $R_1(a,b)$ iff $(a+b)$ is even over the set of integers $R_2(a,b)$ iff $(a+b)$ is odd over the set of integers $R_3(a,b)$ iff $ab > 0$ over the set of non-zero rational numbers $R_4(a,b)$ iff $|a-b| \leq 2$ over the set of natural numbers Which of the following statements is correct?

A. R_1 and R_2 are equivalence relations, R_3 and R_4 are not

B. R_1 and R_3 are equivalence relations, R_2 and R_4 are not

C. R_1 and R_4 are equivalence relations, R_2 and R_3 are not

D. R_1, R_2, R_3 and R_4 are all equivalence relations

23. **[GATE 2001 — 2 Mark]**

Consider the following statements:
S_1: There exist infinite sets A, B, C such that $A \cap (B \cap C)$ is finite.
S_2: There exist two irrational numbers x and y such that $(x + y)$ is rational.
Which of the following is true about S_1 and S_2?

A. Only S_1 is correct

B. Only S_2 is correct

C. Both S_1 and S_2 are correct

D. None of S_1 and S_2 is correct